1. Gramática

No terminales = Δ = {A, B, C} terminales = {+, -, m, f, t}

A -> + A B | +, B -> C - B | mA, C -> f | t

2. Gramática Aumentada

1. A' -> A 2. A -> + A B 3. A -> + 4. B -> C - B

5. B -> m A 6. C -> f 7. C -> t

3. Como No hay producciones con ε, tenemos que:

FIRST(A) = {+}, FIRST(B) = {f, t, m}, FIRST(C) = {f, t}

4. AUTOMATA LR(1)

I_0 = Closure([A' -> .A, $]) = { [A' -> .A, $],
[A -> . + A B, $], [A -> . +, $] }

I_1 = Goto (I_0, A) = {[A' -> A., $]} FIRST (B$) = FIRST(B)

I_2 = Goto(I_0, +) = {[A -> + . A B, $], [A -> + ., $] {}U(+)

(+) = {[A -> . + AB, f], [A -> . + AB, t], [A -> . + AB, m], [A -> . +, f],
[A -> . +, t], [A -> . +, m]} FIRST(-B$) = {-}

I_3 = Goto(I_2, A) = { [A -> + A . B, $], [B -> . C - B, $]
[B -> . m A, $], [C -> . f, -], [C -> . t, -] }

I_4 = Goto(I_2, +) = { [A -> + . A B, f], [A -> + . AB, t],
[A -> + . AB, m], [A -> + ., f], [A -> + ., t], [A -> + ., m]}U(+)

I_5 = Goto(I_3, B) = {[A -> + A B., $]}

I_6 = Goto(I_3, C) = {[B -> C. - B, $]}

$FIRST(\epsilon \$) = \{\$\}$

✓ $I_7 = Goto(I_3, m) = \{[B \to m \cdot A, \$], [A \to \cdot +AB, \$], [A \to \cdot t, \$]\}$

✓ $I_8 = Goto(I_3, f) = \{[C \to f \cdot, -]\}$

✓ $I_9 = Goto(I_3, t) = \{[C \to t \cdot, -]\}$

✓ $I_{10} = Goto(I_4, A) = \{[A \to +A \cdot B, f], [A \to +A \cdot B, t], [B \to \cdot C-B, m], [A \to +A \cdot B, m], [B \to \cdot C-B, f], [B \to \cdot C-B, t], [B \to \cdot mA, f], [B \to \cdot mA, t], [B \to \cdot mA, m], [C \to \cdot f, -], [C \to \cdot t, -]\}$

$Goto(I_4, +) = I_4$ $FIRST(\epsilon \$) = \{\$\}$

✓ $I_{11} = Goto(I_6, -) = \{[B \to C- \cdot B, \$], [B \to \cdot C-B, \$], [B \to \cdot mA, \$], [C \to \cdot f, -], [C \to \cdot t, -]\}$

✓ $I_{12} = Goto(I_7, A) = \{[B \to mA \cdot, \$]\}$

$Goto(I_7, +) = I_2$

✓ $I_{13} = Goto(I_{10}, B) = \{[A \to +AB \cdot, f], [A \to +AB \cdot, t], [A \to +AB \cdot, m]\}$

✓ $I_{14} = Goto(I_{10}, C) = \{[B \to C \cdot -B, f], [B \to C \cdot -B, t], [B \to C \cdot -B, m]\}$

✓ $I_{15} = Goto(I_{10}, m) = \{[B \to m \cdot A, f], [B \to m \cdot A, t], [B \to m \cdot A, m]\} \cup (+)$

$Goto(I_{10}, f) = I_8$, $Goto(I_{10}, t) = I_9$

$I_{16} = Goto(I_{11}, B) = \{[B \to C-B \cdot, \$]\}$

$Goto(I_{11}, C) = I_6$, $Goto(I_{11}, m) = I_7$,

$Goto(I_{11}, f) = I_8$, $Goto(I_{11}, t) = I_9$

$I_{17} = Goto(I_{14}, -) = \{[B \rightarrow C - .B, f], [B \rightarrow C - .B, t], [B \rightarrow C - .B, m], [B \rightarrow .C - B, f], [B \rightarrow .C - B, t], [B \rightarrow .C - B, m], [B \rightarrow .mA, f], [B \rightarrow .mA, t], [B \rightarrow .mA, m], [C \rightarrow .f, -], [C \rightarrow .t, -]\}$

$I_{18} = Goto(I_{15}, A) = \{[B \rightarrow mA., f], [B \rightarrow mA., t], [B \rightarrow mA., m]\}$

$Goto(I_{15}, t) = I_4$

$I_{19} = Goto(I_{17}, B) = \{[B \rightarrow C - B., f], [B \rightarrow C - B., t], [B \rightarrow C - B., m]\}$

$Goto(I_{17}, C) = I_{14}$, $Goto(I_{17}, m) = I_{15}$

$Goto(I_{17}, f) = I_8$, $Goto(I_{17}, t) = I_9$

5º AUTOMATA LALR(1)

$I_0' = Closure([A' \to .A, \$]) = \{[A' \to .A, \$], [A \to .+AB, \$], [A \to .+, \$]\}$

$I_1' = Goto(I_0', A) = \{[A' \to A., \$]\}$

$I_2' = Goto(I_0', +) = I_2 \cup I_4 = \{[A \to +.AB, \$/f/t/m], [A \to +., \$/f/t/m], [A \to .+AB, f/t/m], [A \to .+, f/t/m]\}$

$I_3' = Goto(I_2', A) = I_3 \cup I_{10} = \{[A \to +A.B, \$/f/t/m], [B \to .C-B, \$/f/t/m], [B \to .mA, \$/f/t/m], [C \to .f, -], [C \to .t, -]\}$

$Goto(I_2', +) = I_2'$

$I_4' = Goto(I_3', B) = I_5 \cup I_{13} = \{[A \to +AB., \$/f/t/m]\}$

$I_5' = Goto(I_3', C) = I_6 \cup I_{14} = \{[B \to C.-B, \$/f/t/m]\}$

$I_6' = Goto(I_3', m) = I_7 \cup I_{15} = \{[B \to m.A, \$/f/t/m], [A \to .+AB, \$/f/t/m], [A \to .+, \$/f/t/m]\}$

$I_7' = Goto(I_3', f) = \{[C \to f., -]\} = I_8$

$I_8' = Goto(I_3', t) = \{[C \to t., -]\} = I_9$

$I_9' = Goto(I_5', -) = I_{11} \cup I_{17} = \{[B \to C-.B, \$/f/t/m], [B \to .C-B, f/t/m/\$], [B \to .mA, \$/f/t/m], [C \to .f, -], [C \to .t, -]\}$

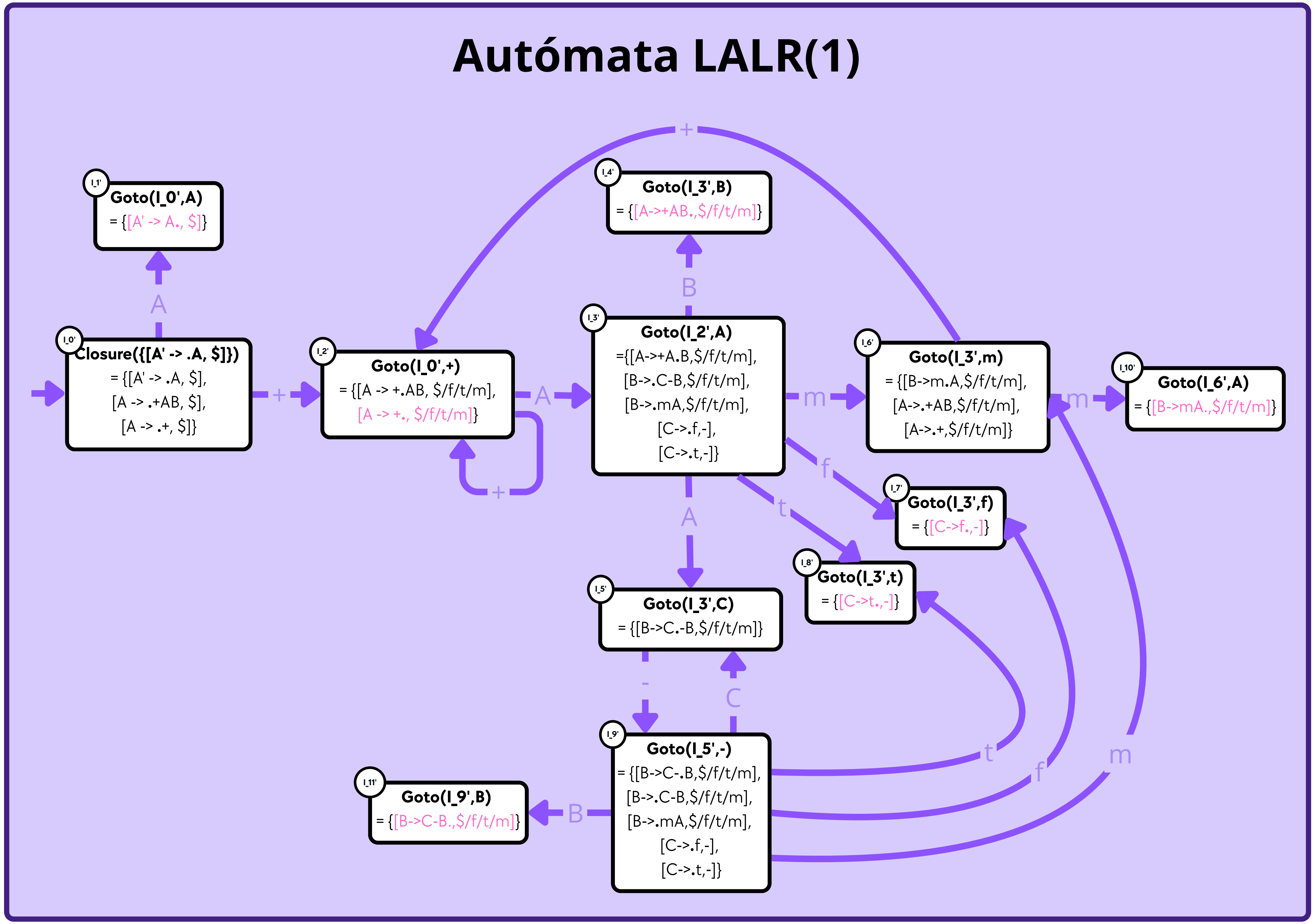
$I_{10}' = Goto(I_6', A) = I_{12} \cup I_{18} = \{[B \to mA., \$/f/t/m]\}$

$Goto(I_6', +) = I_2'$

$I_{11}' = Goto(I_9', B) = I_{16} \cup I_{19} = \{[B \to C-B., \$/f/t/m]\}$

$Goto(I_9', C) = I_5'$, $Goto(I_9', m) = I_6'$

$Goto(I_9', f) = I_7'$, $Goto(I_9', t) = I_8'$



Cadenas de Prueba

Correctas

- +m+
- +m+m+
- +f-m+

Incorrectas

- +f+
- +mf
- ++m